

Grade 6 Accelerated
Unit 10
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

The Customer Service Manager at a call center collected data on the number of minutes a customer was on hold before speaking to a representative. Her data is shown.

8	12	11	7	8	10	9	35	9	10	14
13	4	15	12	10	21	6	5	15	13	11

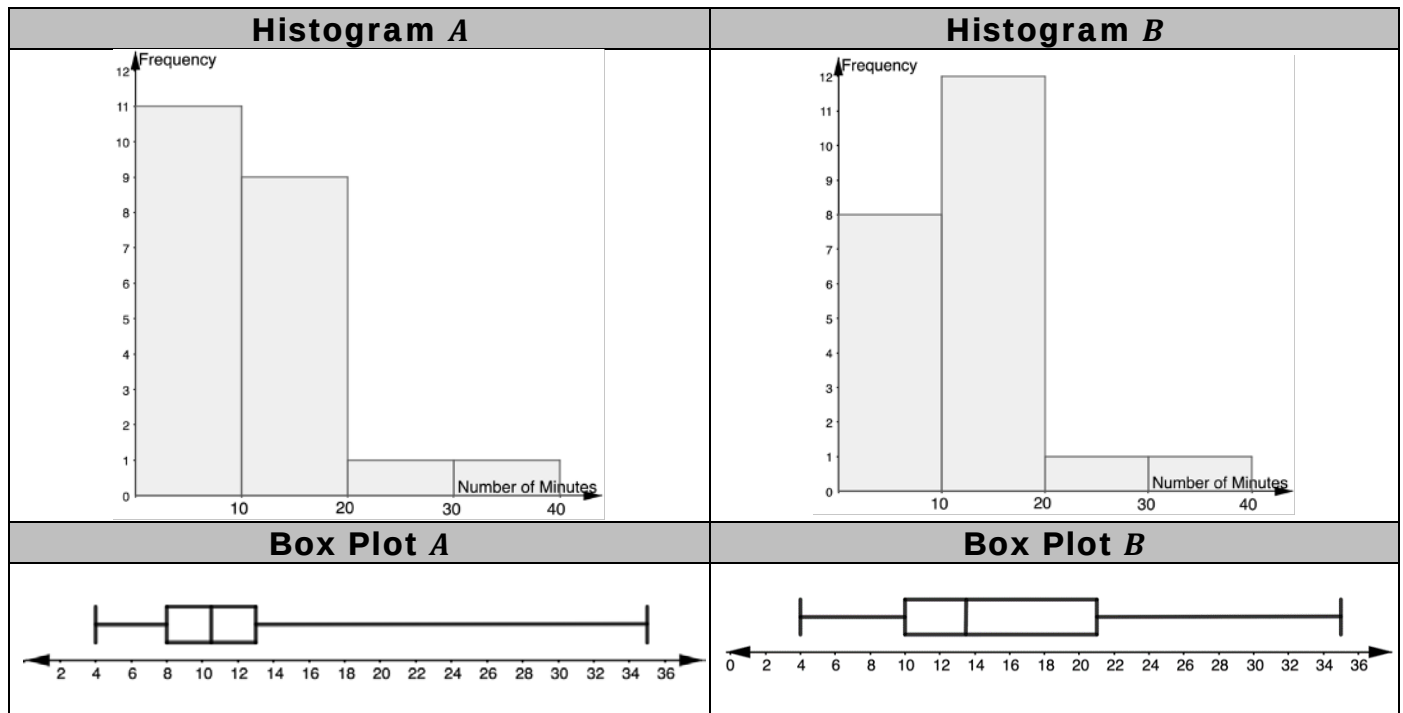
1. Create a frequency table of the data.

Number of Minutes	Frequency
0 – 10	8
10 – 20	12
20 – 30	1
30 – 40	1

2. Determine the five-number summary of the data set.

Min: 4
Q₁: 8
Median: 10.5
Q₃: 13
Max: 35

Two histograms and two box plots are shown.



3. Explain which histogram better represents the data.
Histogram B better matches my frequency table.
4. Explain which box plot better represents the data.
Box Plot A has the correct medians and quartiles.
5. Which graph, box plot or histogram, is better for displaying this data? Explain your reasoning.
Explanations may vary.
I think the histogram gives a better idea of the length of time customers are on hold most often.

Four data displays and four data sets are shown.

Data Display 1	Data Display 2
Data Display 3	Data Display 4
Data Set A	Data Set B
0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 3, 10	0, 3, 4.5, 5, 6, 6.5, 7, 7, 7.5, 8, 9, 9.5
Data Set C	Data Set D
0, 0, 1, 1, 1, 2, 2, 2, 2, 3, 4, 6	0, 2, 4, 6, 6, 7, 7, 7, 7, 8, 8, 8

Match each Data Display to a Data Set.

6. Display 1: **B**
7. Display 2: **C**
8. Display 3: **D**
9. Display 4: **A**
10. Explain how you matched the data set that was represented by Data Display 3.
Explanations may vary.
The range for Display 3 is 0 to 8 and D is the only data set with the same range.